HENRIQUE K. MIYAMOTO

Postdoctoral Researcher · Information Theory and Applications

Flux 7.064, Groene Loper 19, 5612 AP Eindhoven, The Netherlands
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EDUCATION

Université Paris-Saclay, France 2022–2026
Ph.D. in Information Theory
Thesis: Universal Decoding through Randomised Decoding
Committee: Prof. Wigger, Prof. Lapidoth, Dr. Joudeh and Prof. Guillén i Fàbregas

Universidade Estadual de Campinas, Brazil 2021–2022
M.Sc. in Applied Mathematics
Dissertation: Geometry, Statistics and Applications to Communications and Learning

CentraleSupélec, France 2018–2021
Diplôme d'Ingénieur (M.Sc.)
Major: Applied Mathematics

Universidade Estadual de Campinas, Brazil 2015–2021
B.Sc. in Electrical Engineering
Class rank: 1st/74, GPA: 9.196/10

RESEARCH INTERESTS

Information theory and its applications (MSC2020 94A and 94B).

RESEARCH EXPERIENCE

Eindhoven University of Technology (TU/e), The Netherlands 2026–present
Postdoc researcher
Host: Dr. Hamdi Joudeh

L2S, CNRS/CentraleSupélec/Université Paris–Saclay, France 2022–2026
Doctoral researcher
Advisor: Prof. Sheng Yang

IMECC, Universidade Estadual de Campinas, Brazil 2021–2022
Master's researcher
Advisor: Prof. Sueli I. R. Costa

Huawei Technologies, France 2020–2020
Research intern
Supervisors: Dr. Ingmar Land, Dr. Apostolos Desteounis and Prof. Jean-Claude Belfiore

L2S, CNRS/CentraleSupélec/Université Paris–Saclay, France 2019–2020
Student researcher
Supervisor: Prof. Sheng Yang

IMECC, Universidade Estadual de Campinas, Brazil 2016–2018
Undergradute student researcher
Supervisors: Prof. Henrique N. Sá Earp and Prof. Sueli I. R. Costa

TEACHING TRAINING

Université Paris-Saclay, France 2022–2026
Certificate "Teaching in Higher Education" (*Parcours "Enseigner dans le Supérieur"*)

- This program certifies training in different teaching skills, and teaching experience of at least 96h, done in parallel to the preparation of the Ph.D. degree.

TEACHING EXPERIENCE

CentraleSupélec, France
Graduate teaching assistant 2022–2025

- Information theory
- Signal processing
- Communication theory
- Digital signal processing
- Wireless communications

Universidade Estadual de Campinas, Brazil
Graduate teaching assistant 2022–2022

- Principles of communications I

Undegraduate teaching assistant 2021–2021

- Principles of communications I

PREPRINTS

[Z3] H. K. Miyamoto, R. Combes and S. Yang, "Mismatched exponents for deterministic and randomised noise-guessing decoding", *2606.26954*, 2026. doi: 10.48550/arXiv.2606.26954.

[Z2] H. K. Miyamoto and S. Yang, "Error exponents for randomised list decoding", *arXiv:2601.09519*, 2026. doi: 10.48550/arXiv.2601.09519.

[Z1] H. K. Miyamoto and S. Yang, "Universal decoding over finite-state additive channels via noise guessing", *arXiv:2501.12971*, 2025. doi: 10.48550/arXiv.2501.12971.

JOURNAL PAPERS

[J4] H. K. Miyamoto, F. C. C. Meneghetti, J. Pinele and S. I. R. Costa, "On closed-form expressions for the Fisher–Rao distance", *Information Geometry*, vol. 7, pp. 311–354, Nov. 2024. doi: 10.1007/s41884-024-00143-2.

[J3] H. K. Miyamoto, F. C. C. Meneghetti and S. I. R. Costa, "The Fisher-Rao loss for learning under label noise", *Information Geometry*, vol. 6, pp. 107–126, Jun. 2023. doi: 10.1007/s41884-022-00076-8.

[J2] H. K. Miyamoto and S. Yang, "Context-tree-based lossy compression and its application to CSI representation", *IEEE Transactions on Communications*, vol. 70, no. 7, pp. 4417–4428, Jul. 2022. doi: 10.1109/TCOMM.2022.3173002.

[J1] H. K. Miyamoto, S. I. R. Costa and H. N. Sá Earp, "Constructive spherical codes by Hopf foliations", *IEEE Transactions on Information Theory*, vol. 67, no. 12, pp. 7925–7939, Dec. 2021. doi: 10.1109/TIT.2021.3114094.

CONFERENCE PAPERS

[C9] R. Zhang, H. Miyamoto, R. Combes and S. Yang, "An efficient synchronization scheme for distributed bandits", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1–6, Guangzhou, 2026. doi: 10.1109/ISIT62367.2026.11654092.

[C8] H. K. Miyamoto and S. Yang, "On α-tilted noise guessing decoding", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1–6, Guangzhou, 2026. doi: 10.1109/ISIT62367.2026.11653642.

[C7] H. K. Miyamoto and S. Yang, "Error exponents for randomised list decoding", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1–6, Guangzhou, 2026. doi: 10.1109/ISIT62367.2026.11654089.

[C6] H. K. Miyamoto and S. Yang, "On universal decoding over discrete additive channels by noise guessing", *IEEE Information Theory Workshop (ITW)*, pp. 1–6, Sydney, 2025. doi: 10.1109/ITW62417.2025.11240417.

[C5] H. K. Miyamoto and S. Yang "On universal decoding over memoryless channels with the Krichevsky–Trofimov estimator", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1498-1503, Athens, 2024. doi: 10.1109/ISIT57864.2024.10619414.

[C4] F. C. C. Meneghetti, H. K. Miyamoto, S. I. R. Costa and M. H. M. Costa, "Revisiting lattice tiling decomposition and dithered quantisation", *Geometric Science of Information (GSI)*, Saint-Malo, 2023. In: F. Nielsen and F. Barbaresco (eds.), *GSI 2023*, LNCS 14071, pp. 318–327, Springer, 2023. doi: 10.1007/978-3-031-38271-0_319.

[C3] F. C. C. Meneghetti, H. K. Miyamoto and S. I. R. Costa, "Information properties of a random variable decomposition through lattices", *41st International Conference on Bayesian and Maximum Entropy Methods in Science and Engineering (MaxEnt)*, Paris, 2022. In: *Physical Sciences Forum*, vol. 5, no. 1, 2022. doi: 10.3390/psf2022005019.

[C2] H. K. Miyamoto and S. Yang, "A CSI compression scheme using context trees", *International Zurich Seminar on Information and Communication (IZS)*, pp. 24–28, Zurich, 2022. doi: 10.3929/ethz-b-000535273.

[C1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Constructive spherical codes in 2^k dimensions", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1612–1616, Paris, 2019. doi: 10.1109/ISIT.2019.8849464.

DOMESTIC CONFERENCE PAPERS

[D3] H. K. Miyamoto, "Geometria, estatística e aplicações a comunicações e aprendizado"[1], *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, vol. 10, no. 1, 2023. doi: 10.5540/03.2023.010.01.0004 (in Portuguese).

[D2] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Construção de códigos esféricos usando a fibração de Hopf", *Jornada Nacional de Iniciação Científica (JNIC), 70ª Reunião Anual da SBPC*, Maceió, 2018. ISSN: 2176-1221 (in Portuguese).

[D1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Construction of spherical codes using the Hopf fibration", *XXV Congresso de Iniciação Científica da Unicamp*, Campinas, 2017. doi: 10.19146/pibic-2017-78809.

PATENT APPLICATIONS

[P1] I. Land, H. K. Miyamoto, A. Destounis and J.-C. Belfiore, "Transfer Learning between Neural Networks", WO2022207097 (2022), CN117121020 (2023), EP4309079 (2024).

[1] This is an extended abstract of [T1], published on the occasion of the *Clóvis Caesar Gonzaga Award* (see Awards).

THESES

[T2] H. K. Miyamoto, "Universal Decoding by Randomised Decoding", *Ph.D. thesis*, Université Paris-Saclay, Gif-sur-Yvette, 2026. hal: tel-05749367v1.

[T1] H. K. Miyamoto, "Geometria, Estatística e Aplicações a Comunicações e Aprendizado" (Geometry, Statistics and Applications to Communications and Learning), *M.Sc. dissertation*, Universidade Estadual de Campinas, Campinas, 2022. hdl: 20.500.12733/6633.

SELECTED PRESENTATIONS (WITHOUT PROCEEDINGS)

2nd France PhD Workshop on Information Theory, *oral presentation*, Gif-sur-Yvette, France, 2026.

IEEE European School on Information Theory (ESIT), *poster presentation*, Eindhoven, The Netherlands, 2024.

1st France PhD Workshop on Information Theory, *poster presentation*, Palaiseau, France, 2024.

IEEE European School on Information Theory (ESIT), *poster presentation*, Bristol, UK, 2023.

International Conference on Information Geometry for Data Science (IG4DS), *oral presentation*, Hamburg, Germany (virtual), 2022.

International Congress of Mathematicians (ICM), *oral presentation*, Rio de Janeiro, Brazil, 2018.

Latin American Week on Coding and Information (LAWCI), *poster presentation*, Campinas, Brazil, 2018.

Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), *oral presentation*, Curitiba, Brazil, 2017.

AWARDS

1st place in **Clóvis Caesar Gonzaga Award** for master's dissertations in applied and computational mathematics, SBMAC, 2023.

Merit Award in Jornada Nacional de Iniciação Científica (JNIC), SBPC, 2018.

2nd place in **Beatriz Neves Award** for undergraduate research in applied and computational mathematics, SBMAC, 2018.

Academic Excellence in Undergraduate Work in Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), category Geometry and Topology, PET Matemática/UFPR, 2017.

Scientific Merit in XXV Congresso de Iniciação Científica da Unicamp, PRP/Unicamp, 2017.

Merit Award for the composition produced in Unicamp entrance exam , COMVEST/Unicamp, 2015.

SCHOLARSHIPS

2021–2022, *Master's Degree Scholarship*, São Paulo Research Foundation (FAPESP), 2021/04516-8.

2018–2020, *Eiffel Excellence Scholarship*, French Ministry for Europe and Foreign Affairs.

2016–2018, *Scientific Initiation Scholarship*, São Paulo Research Foundation (FAPESP), 2016/05126-0.

2013–2013, *Junior Scientific Initiation Scholarship*, Brazilian National Council for Scientific and Technological Development (CNPq).

PROFESSIONAL SERVICE AND ENGAGEMENT

Scientific Societies Membership

2025–present, *Société Mathématique de France (SMF)*, France.

2025–present, *Société de Mathématiques Appliquées et Industrielles (SMAI)*, France.

2020–present, *Sociedade Brasileira para o Progresso da Ciência (SBPC)*, Brazil.

2018–2023, *Sociedade Brasileira de Matemática Aplicada e Computacional (SBMAC)*, Brazil.

2018–present, *Institute of Electrical and Electronics Engineers (IEEE)*, USA.

- **Information Theory Society**, Communications Society.

Reviewing

Journals

- IEEE Transactions on Information Theory
- IEEE Communications Letters
- Information Geometry (Springer)
- Journal of Communication and Information Systems

Conferences

- IEEE International Symposium on Information Theory (ISIT 2023, 2024, 2025, 2026)
- Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT 2023)

Committees and Representative Positions

2023–2025, *doctoral and postdoc researchers representative*, Laboratory of Signals and Systems (L2S).

2023, *organising committee*, Encontro de Códigos, Reticulados e Informação, IMECC, Unicamp.

2018–2019, *student representative*, Departamento de Comunicações, FEEC, Unicamp.

LANGUAGES

Portuguese	native
English	advanced (Cambridge CAE)
French	advanced
Spanish	intermediate
Japanese	elementary (JLPT N4)

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